



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

ARTIFICIAL INTELLIGENCE
SECJ3553
Project Proposal

Theme: Intelligent Volunteer Matching

Team name: Visionary.AI

Team Members:

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AI Project Proposal Template

1. Project Title

Intelligent Volunteer Matching

2. Project Overview

Background & Rationale:

The inefficiency that traditional volunteer recruitment causes by wasting so much time on filtering and searching the resumes of people who want to do charity work. It wastes so much of the recruiter's time and sometimes ends up putting people in positions that they are overqualified for so it ends up wasting their potential and time too.

Description of AI Solution:

Our AI solution aims to match jobs with people who are qualified for these jobs based on mutual attributes of these jobs and the people who want to volunteer. For example If a person is a good graphic designer and there is a job that requires designing or making a logo or a poster. The system will match that the person is a designer for visuals and graphics and match that with the requirement to design and make a poster.

Goal of AI Solution:

The AI goal in this project is to automate the filter and matching that each job offered needs. It is intended to save time where all resumes are filtered each time for a single job into matching each candidate and recommending the job that matches their skills set the best.

Expected Impact / Value:

This solution is expected to greatly increase the efficiency in choosing the correct candidates for each job to raise the quality of the delivered work. And for the volunteers side it is expected to help them gain more experience and get more benefit from the time they give in volunteering by upgrading their skills and knowledge in their respective field or profession.

4. Design Thinking – Empathize

Target Users / Stakeholders:

1- Volunteer Coordinators :

- Manage recruitments for events and programs.
- Need fast, and accurate ways to find qualified and reliable volunteers.
- often spend too much manually going through applications and still ends up unsure if the volunteers will actually be suitable or dependable.

2- Skilled Volunteers like : (Designers, Developers, Doctors, etc...)

- People that want to use their professional skills to make contributions.
- Often feel that they are not utilized when matched with unrelated roles.

3- General Volunteers like : (Students, Community Members, etc)

- People that want to join causes that match their passions.
- Might have limited time or specific availability, making it hard to find suitable opportunities for them.

User Needs & Pain Points:

Summarize key insights and emotions captured from users.

User Type	Needs	Pain Points
Volunteer Coordinators	- Find volunteers with specific skills and availability. - Reduce time spent manually filtering applications.	- Overwhelmed by too many irrelevant applications. - Difficulty verifying skill fit and reliability.

Skilled Volunteers	- Opportunities that match their expertise. - Meaningful, skill-based impact.	- Current platforms don't recognize or value their specific skills. - Feel their time and ability are wasted.
General Volunteers	- Easy access to opportunities matching interests and schedules. - Flexible and transparent volunteer matching.	- Frustrated by irrelevant results and scheduling conflicts. - Lose motivation due to poor matches.

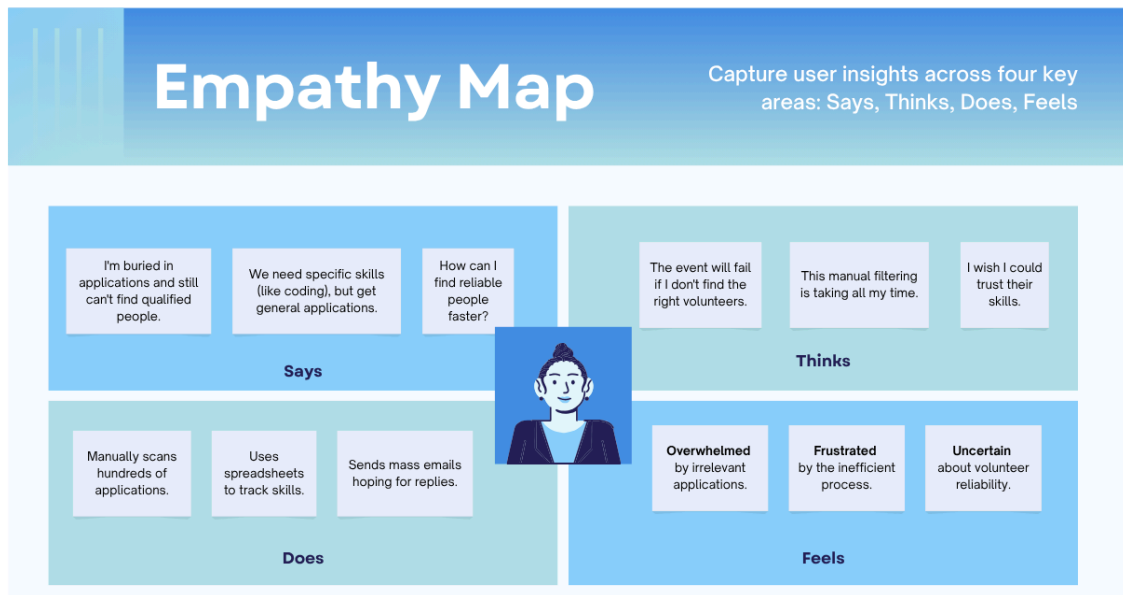


Figure 1.0 Sample for Empathize with Diagram

5. Design Thinking – Define

Problem Statement:

Non-profit organizations often struggle to find volunteers who *genuinely match* the skills, interests, and schedules required for their activities. As a result, coordinators waste time filtering unsuitable applicants, skilled volunteers feel underutilized, and many general volunteers lose motivation because opportunities do not align with their availability or passions. This mismatch causes high no-show rates, low engagement, and reduced impact on the community.

There is a need for an intelligent, reliable, and personalized volunteer-opportunity matching system that ensures the right volunteers are connected to the right causes.

Key Requirements for the AI Solution:

ID	Requirement	Description
FR1	Volunteer Profile Processing	The system must collect volunteer data (skills, interests, availability, location) through forms or uploaded resumes.
FR2	AI Skill & Interest	The system must automatically extract and match skills and interests from volunteer forms.
FR3	Opportunity Requirement Input	Organizations must be able to create volunteer opportunities with required skills, time slots, location, and cause category.
FR4	AI Matching Algorithm	The system must compute a match score between each volunteer and each opportunity based on skill fit, interest alignment, availability, and location proximity.
FR5	Personalized Recommendations	Volunteers must receive a ranked list of opportunities that best match their profile.
FR6	Coordinator Dashboard	Coordinators must be able to view the top

		recommended volunteers for each posted opportunity.
FR7	Notification System	The system should notify volunteers and coordinators when new high-match opportunities or volunteers appear.

9. Task and Responsibility

Task	Description	PIC
Describe the process of Empathize in DT	Problem and pain, user/stakeholder and the goal are clearly described. Who is my user? What matters to them?	Mohamed Ali Mohamed Ali and Goh Jiale
Describe the process of Define in DT	Covers user/stakeholder needs in-depth with details and examples. What are their needs?	Abdelrahman Osama Said
The goal of AI solution	AI solutions show a large amount of original thought. Ideas are creative and inventive.	Ling Yu Qian and all member (discuss and think on solution)
Name AI solution	No specific description — space to write name of AI solution.	Ling Yu Qian

10. References / Supporting Materials

1. Dam, R. F., & Teo, Y. S. (2025, August 17). *Empathy map – Why and how to use it*. Interaction Design Foundation.
<https://www.interaction-design.org/literature/article/empathy-map-why-and-how-to-use-it>
2. National Institutes of Health. (2024, November 25). *AI algorithm successfully matches potential volunteers to clinical trials*. ScienceDaily. Retrieved November 4, 2025, from www.sciencedaily.com/releases/2024/11/241118125819.htm
3. West, D. M. (2018, June 5). *Artificial intelligence in non-profit organizations*. Brookings Institution.
<https://www.brookings.edu/articles/artificial-intelligence-in-non-profit-organizations/>
4. VolunteerMatch. (2025). *Connecting volunteers with nonprofit organizations*. Retrieved November 4, 2025, from <https://www.volunteermatch.org/>